

Safety

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1 Problem Formulation

In multi-robot navigation, we would like to avoid robots colliding into each other. “Control barrier functions” (CBFs) are a way to look at safety. We will assume that we can represent the state of our system as a subset of $\mathbb{R}^n$ for some positive integer n . These can, for example, be the positions and velocities of all robots in the system. We will also assume that we can model our safe set as

$$C = \{x \in \mathbb{R}^n \mid h(x) \geq 0\} \quad (1)$$

where x is a state of our system, and $h : \mathbb{R}^n \rightarrow \mathbb{R}$ is a so-called “barrier function”. Furthermore, our dynamics will be of the form

$$\dot{x} = f(x) + g(x) \cdot u \quad (2)$$

where u is our m -dimensional control [often acceleration], $g : \mathbb{R}^n \rightarrow \mathbb{R}^{n \times m}$ is how the control gets applied, and $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is a drift term that we do not have control over. This is a “control affine” system and appears often in real world robots.

2 Lyapunov Stability

Consider a system consisting of the state x , which has an equilibrium at $t = 0$, and a function $V : \mathbb{R}^n \rightarrow \mathbb{R}$. Broadly, this system is Lyapunov stable if the states that start out near the equilibrium state stay near it. Our function V here represents the potential function.

More formally, this system is Lyapunov stable if both of the following conditions hold:

1. $V(x) > 0$ for all $x \neq 0$, and $V(0) = 0$
2. $\dot{V}(x) \leq 0$ for all x , and $\dot{V}(0) = 0$

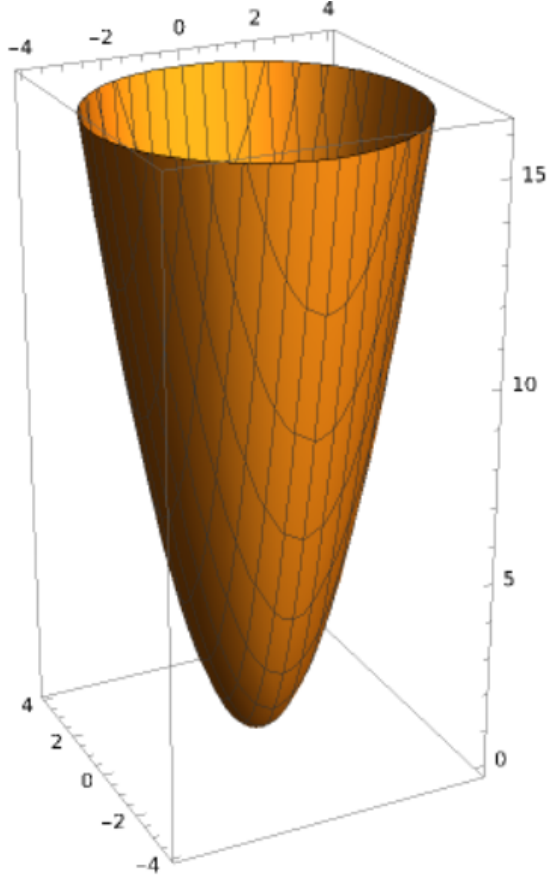
This indicates that the states x are driven toward $x = 0$, as this has $V(x) = 0$. We can think of V as a bowl-shaped function. The intuition here is that we are going toward the center of the bowl. When V represents the potential, stability is maximized at the minimum energy, which happens with $V(x) = 0$. Figure 1 shows examples illustrating the Lyapunov function and dynamics.

3 Lyapunov Stability for Safety

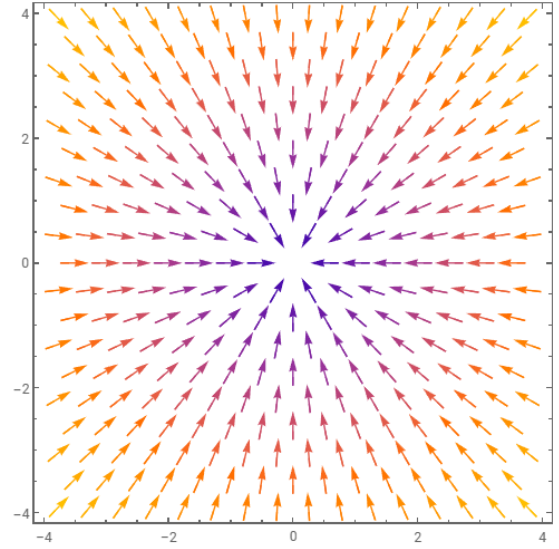
Suppose we require V to be always decreasing as a function of time (which we can think of as always getting safer). We have the constraint

$$\begin{aligned} \dot{V}(x) &= \frac{dV}{dt} = \frac{\partial V}{\partial x} \frac{\partial x}{\partial t} = \frac{\partial V}{\partial x} \dot{x} = \frac{\partial V}{\partial x} [f(x) + g(x) \cdot u] \\ &= \frac{\partial V}{\partial x} f(x) + \frac{\partial V}{\partial x} g(x) \cdot u \leq 0 \end{aligned} \quad (3)$$

This is a linear constraint on u . We want $\arg \min_{u_t^i} \|u_t^i - \tilde{u}_t^i\|^2$ such that $\dot{V} \leq 0$ holds. Here $\tilde{u}_t^i$ is what the original control (ignoring the safety consideration) by robot i at time t would be, and we are looking for the closest u_t^i such that $\dot{V}(x) \leq 0$. This is a quadratic objective with a linear constraint, and so can be used by real robots in real-time, as it is very fast to calculate.



(a) Example of a Lyapunov function V .



(b) Example dynamics with Lyapunov stability. Note how all points go toward the center.

Figure 1: Visualization of Lyapunov stability and dynamics.

4 Barrier Certificate

Suppose we wanted safety in a similar barrier certificate that can be incorporated as a constraint. There is a simple way to derive a similar condition for safety.

We take our Lyapunov function, (additively) invert it, and then add to it. This gives us our barrier function. That is, the barrier function is of the form $h(x) = c - V(x)$, where c is a positive constant, and $V(x)$ is a Lyapunov function. This then creates a region $\mathcal{C}$ where $h(x) > 0$ for all $x \in \mathcal{C}$, $h(x) = 0$ for all $x \in \partial\mathcal{C}$, and $h(x) < 0$ otherwise.

As with the Lyapunov function, we have

$$\dot{h} = \frac{\partial h}{\partial x} \dot{x} = \frac{\partial h}{\partial x} f(x) + \frac{\partial h}{\partial x} g(x) \cdot u$$

The constraint $\dot{h} > 0$ is too restrictive. Even when we're already in the safe region $\mathcal{C}$, the condition $\dot{h} > 0$ would pull us toward the center of $\mathcal{C}$. Instead, we want to be minimally restrictive. Thus, we say

$$\dot{h} \geq -\alpha h(x)$$

Here, α is positive. When the robot is already safe, $\dot{h}$ is allowed to be negative (since $h(x)$ is positive), so we are allowed to move toward the boundary a bit. But when we're not safe, $\dot{h}$ must be positive, and we'll get pulled toward the safe region. Finally, when we're on the boundary, $\dot{h}$ satisfies $\dot{h} \geq 0$, so we won't become unsafe afterward. As you go from the center of the safe region to the boundary to outside, $\dot{h}$ gets more and more restricted.

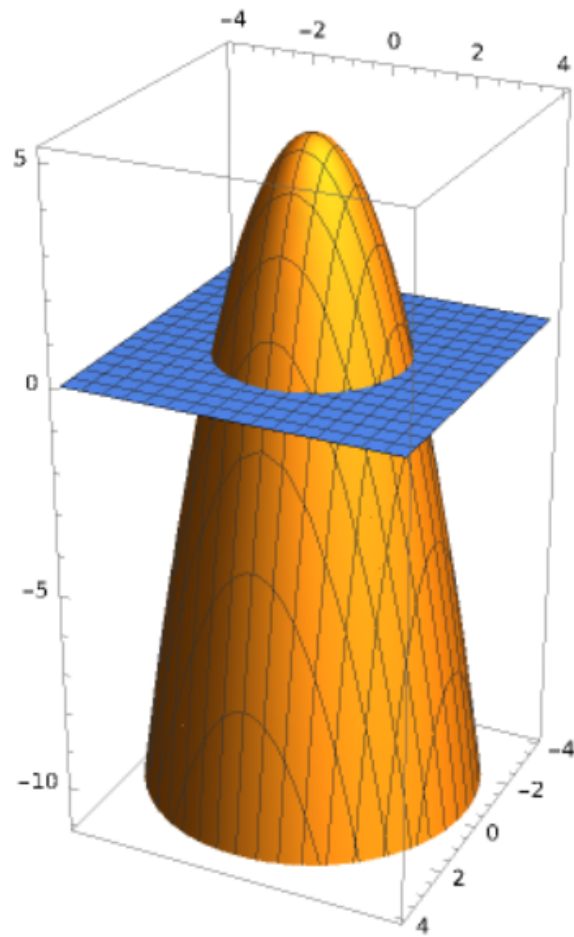


Figure 2: Example barrier function. In the region such that h is above the plane, we have $h(x) > 0$. At the boundary, we have $h(x) = 0$. For other values, we have $h(x) < 0$.

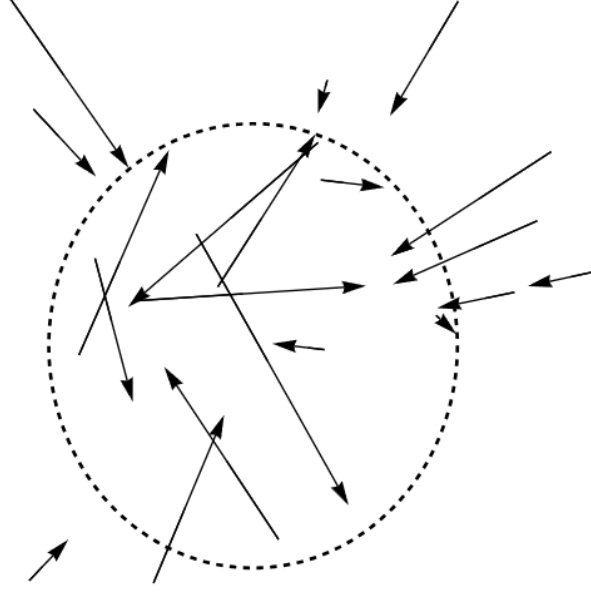


Figure 3: Example showing allowed controls. The dotted boundary indicates the safe region. Outside the safe region, we must move toward it. But inside it, we don’t necessarily have to move toward the center (we just have to remain in the safe region).

5 Safety Barrier Certificates for Heterogeneous Multi-Robot Systems (Wang, Ames & Egerstedt, 2016)

Abstract—This paper presents a formal framework for collision avoidance in multi-robot systems, wherein an existing controller is modified in a minimally invasive fashion to ensure safety. We build this framework through the use of control barrier functions (CBFs) which guarantee forward invariance of a safe set; these yield safety barrier certificates in the context of heterogeneous robot dynamics subject to acceleration bounds. Moreover, safety barrier certificates are extended to a distributed control framework, wherein neighboring agent dynamics are unknown, through local parameter identification. The end result is an optimization-based controller that formally guarantees collision free behavior in heterogeneous multi-agent systems by minimally modifying the desired controller via safety barrier constraints. This formal result is verified in simulation on a multi-robot system consisting of both “sluggish” and “agile” robots.

[1] introduces a way to use control barrier functions to guarantee safety. If we start off in this “safe set”, we are guaranteed to always remain in this set. The same approach can be used in a distributed control setting, where each robot needs to learn the neighboring robots’ parameters. And it does so by minimally altering the original controller, which is an often-desired secondary objective.

6 Control Barrier Constraint for Multi-Robot Navigation

Suppose we can write the state of the i th robot at time t as $x_t^i = [p_t^i, v_t^i]$, where p_t^i and v_t^i are both elements of $\mathbb{R}^2$ representing the position and velocity, respectively. Moreover, write as from the first lecture, the observation as

$$o_t^i(x_t^i) = \{x_t^i\} \cup \{x_t^j \mid j \in N(x_t^i)\}.$$

$N(x_t^i)$ here represents the neighborhood of robots that are able to be observed.

Our double integrator dynamics for $\dot{x} = f(x) + g(x) \cdot u$ are given as

$$\begin{pmatrix} \dot{p}^i \\ \dot{v}^i \end{pmatrix} = \begin{pmatrix} 0 & I_{2 \times 2} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} p^i \\ v^i \end{pmatrix} + \begin{pmatrix} 0 \\ I_{2 \times 2} \end{pmatrix} u^i$$

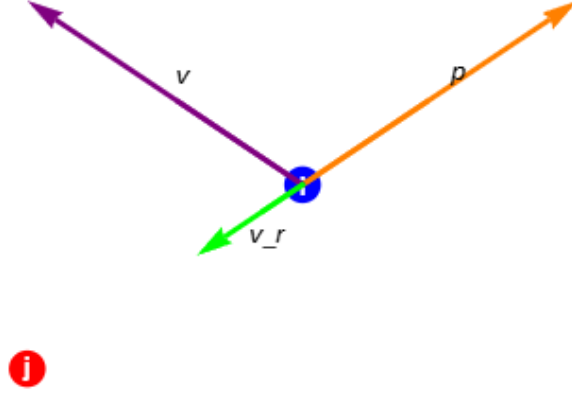


Figure 4: Enter Caption

where u^i is the acceleration of the i th robot. We also have a "safe distance" d that says any robot must be at least d away from all other robots.

For now, let's look at the safety constraints when considering just the i th and j th robot. We essentially want to say that when they apply their maximum braking acceleration, they will not collide (i.e. will be at least d apart). Let $\bar{p} = p_i - p_j$ and $\bar{v} = v_i - v_j$ be the relative position and velocity vectors between these two robots. Let α_i be the maximum braking acceleration of the i th robot. Also define

$$\bar{v}_r = \frac{\bar{p}^T}{\|\bar{p}\|} \bar{v}$$

This is the normal component of $\bar{v}$ that might lead to collision.

The constraint we have is

$$\|\bar{p}\| + \int_{t_0}^{t_0+T} \bar{v}_r(t) dt \geq d$$

But if they both apply maximum braking at t_0 toward the other robot, we have $\bar{v}_r(t_0+t) = \bar{v}_r(t_0) + (\alpha_i + \alpha_j)t$. The constraint then simplifies to

$$\|\bar{p}\| + t\bar{v}_r + (\alpha_i + \alpha_j) \left[\frac{(t_0+t)^2 - t_0^2}{2} \right] \geq d$$

Furthermore, let's say that $t_0 = 0$, and we stop at t_b , the braking time (time to reach 0 relative velocity). Note that t_b is given by $-\frac{\bar{v}_r}{\alpha_i + \alpha_j}$. If we plug in $t_0 = 0$ and $t = -\frac{\bar{v}_r}{\alpha_i + \alpha_j}$, we get

$$\|\bar{p}\| - \frac{\bar{v}_r^2}{2(\alpha_i + \alpha_j)} \geq d$$

If we solve this for $\bar{v}_r$, we get that

$$-\sqrt{R} \leq \bar{v}_r \leq \sqrt{R}$$

where $R = 2(\alpha_i + \alpha_j)[\|\bar{p}\| - d]$. But we only care about what happens when $\bar{v}_r < 0$, since the two robots can only collide if they are moving closer. So our condition becomes

$$\bar{v}_r = \frac{\bar{p}^T}{\|\bar{p}\|} \bar{v} \geq -\sqrt{2(\alpha_i + \alpha_j)[\|\bar{p}\| - d]}.$$

This is equation (6) in [1], where we have

$$h_{ij} = \sqrt{2(\alpha_i + \alpha_j)[\|\bar{p}_{ij}\| - d]} + \frac{\bar{p}_{ij}^T}{\|\bar{p}_{ij}\|} \bar{v}_{ij} \geq 0$$

6.1 Constraint on Control u

In order to find the actual constraint on our control, we first need to find $\dot{h}_{ij}$. Considering the double integrator dynamics for the relative position/velocities, we have

$$\begin{pmatrix} \dot{\bar{p}} \\ \dot{\bar{v}} \end{pmatrix} = \begin{pmatrix} 0 & I_{2 \times 2} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \bar{p} \\ \bar{v} \end{pmatrix} + \begin{pmatrix} 0 \\ I_{2 \times 2} \end{pmatrix} \bar{u}$$

We are trying to calculate $\dot{h} = \frac{\partial h}{\partial x} f(x) + \frac{\partial h}{\partial x} g(x) \cdot u$. The first term is equal to

$$\begin{pmatrix} \frac{\partial h}{\partial \bar{p}} & \frac{\partial h}{\partial \bar{v}} \end{pmatrix} \cdot \begin{pmatrix} 0 & I_{2 \times 2} \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \bar{p} \\ \bar{v} \end{pmatrix} = \frac{\partial h}{\partial \bar{p}} \bar{v}$$

Similarly, the second term is $\frac{\partial h}{\partial \bar{v}} \bar{u}$. So we get

$$\dot{h} = \frac{\partial h}{\partial \bar{p}} \bar{v} + \frac{\partial h}{\partial \bar{v}} \bar{u}$$

If we plug in our earlier expression for h , we get

$$\dot{h} = \frac{(\alpha_i + \alpha_j)}{\sqrt{2(\alpha_i + \alpha_j)(\|\bar{p}\| - d)}} \cdot \frac{\bar{v}^T \bar{p}}{\|\bar{p}\|} - \frac{(\bar{v}^T \bar{p})^2}{\|\bar{p}\|^3} + \frac{\|\bar{v}\|^2}{\|\bar{p}\|} + \frac{\bar{p}^T}{\|\bar{p}\|} \bar{u}$$

We now would want $\dot{h} \geq -\alpha(h)$. [1] sets α not to a constant, but defines $\alpha(x) = \gamma x^3$, where γ is a positive constant. This is due to some theoretical results they obtain, where they require α to satisfy some nice properties (including that it must be strictly increasing and have $\alpha(0) = 0$) in order to get their safety guarantees. If we simplify the resulting expression and write it so $\bar{u}$ appears on one side, we get equation (9) in [1],

$$-\bar{p}^T \bar{u} \leq \gamma h^3 \|\bar{p}\| - \frac{(\bar{v}^T \bar{p})^2}{\|\bar{p}\|^2} + \|\bar{v}\|^2 + \frac{(\alpha_i + \alpha_j) \bar{v}^T \bar{p}}{\sqrt{2(\alpha_i + \alpha_j)(\|\bar{p}\| - d)}}$$

This constraint over all pairs of robots can then be represented as a set of linear constraints on $\bar{u}$.

7 CBFs vs Constrained Trajectory Optimization

Control barrier functions have the advantage that they guarantee safety pointwise at every timestep. That is, even if we analyze at intervals of Δt (instead of truly continuously), they will look in advance to see what will happen in the worst case between $n \cdot \Delta t$ and $(n+1) \cdot \Delta t$. This is unlike constrained trajectory optimization, which generates a safe trajectory using a single “snapshot” of the environment at the beginning, but then the control is frozen. In that Δt time, things can go wrong if we were to use constrained trajectory optimization.

7.1 Example

Let's look at an example. Suppose there is an obstacle at $(5, 5)$ that we want to maintain a distance of r from. Our barrier function h is given by $h(x) = \|p^i - (5, 5)\| - r = \sqrt{(p_x - 5)^2 + (p_y - 5)^2} - r \geq 0$. Our safety filter, which is used for our controls, is of the form $h'(x) + \alpha h(x) \geq 0$, where α is a positive constant. Suppose also that our dynamics is of the form

$$\begin{pmatrix} \dot{p}_x \\ \dot{p}_y \\ \dot{\theta} \end{pmatrix} = \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix} u_v + \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} u_w = \begin{pmatrix} \cos \theta & 0 \\ \sin \theta & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} u_v \\ u_w \end{pmatrix}$$

We can imagine our controls u as controlling how much we accelerate linearly and how much we accelerate rotationally.

Let's go back to our prior formulation of $\dot{h} = \frac{\partial h}{\partial x} f(x) + \frac{\partial h}{\partial x} g(x) \cdot u$. Since $f(x) = 0$, the first term is 0. The second term is given by

$$\frac{\partial h}{\partial p_x} u_v \cos \theta + \frac{\partial h}{\partial p_y} u_v \sin \theta + \frac{\partial h}{\partial \theta} u_w$$

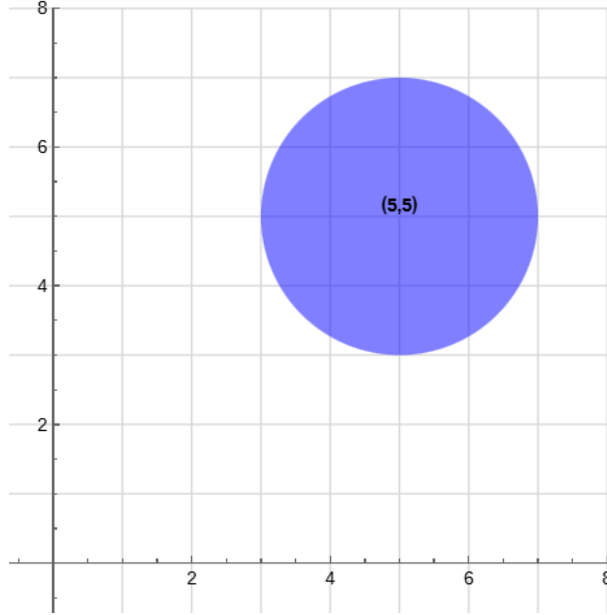


Figure 5: Example with obstacle at (5, 5) and safe distance of 2.

The partial derivatives for this system are relatively easy to find. We have $\frac{\partial h}{\partial p_x} = \frac{p_x - 5}{\sqrt{(p_x - 5)^2 + (p_y - 5)^2}}$, and similarly for p_y . $\frac{\partial h}{\partial \theta}$ is just 0. So we have

$$\dot{h} = \frac{(p_x - 5)u_v \cos \theta + (p_y - 5)u_v \sin \theta}{\sqrt{(p_x - 5)^2 + (p_y - 5)^2}}$$

Notice how this has the dynamics incorporated into it.

Let's instead look at what an MPC does. Assume $r = 2$, which would mean that we have to be at least 2 away from the obstacle at (5, 5). Suppose that our robot starts at (2, 2). The MPC will try to find

$$\min_u \sum_{t=0}^{T-1} \ell(x_t^i, u_t^i)$$

At the beginning, the robot is safe because $\|(5, 5) - (2, 2)\| = \sqrt{18} \geq 2$. The MPC plans a trajectory that stays at least 2 away from the obstacle. Suppose $\Delta t = 0.2$, and suppose it would ideally end up at (3.58, 3.58) [which is still a distance of greater than 2 away from the obstacle]. But between $t = 0$ and $t = 0.2$, a disturbance could cause drift to (3.6, 3.6), which is at a distance of only about 1.98 from the obstacle.

An MPC only enforces safety at $t = 0$, $t = \Delta t$, $t = 2\Delta t$, and so on. But disturbances could happen between these points that cause deviations from the planned trajectory, resulting in unsafe behavior.

8 Safety-Critical Model Predictive Control with Discrete-Time Control Barrier Function (Zeng, Zhang, & Sreenath, 2021)

Abstract—The optimal performance of robotic systems is usually achieved near the limit of state and input bounds. Model predictive control (MPC) is a prevalent strategy to handle these operational constraints, however, safety still remains an open challenge for MPC as it needs to guarantee that the system stays within an invariant set. In order to obtain safe optimal performance in the context of set invariance, we present a safety-critical model predictive control strategy utilizing discrete-time control barrier functions (CBFs), which guarantees system safety and accomplishes optimal performance via model predictive control. We analyze the feasibility and the stability properties of our control design. We verify the properties of our

method on a 2D double integrator model for obstacle avoidance. We also validate the algorithm numerically using a competitive car racing example, where the ego car is able to overtake other racing cars.

[2] introduced a way to get around the safety problem in between the discrete timesteps when using MPC that we saw in the example. They combined MPC with CBFs in order to get MPC-CBF. Prior work had already generalized CBFs into discrete-time systems. But this can be further improved by predicting our future states, and this would result in a better control policy. In their work, the “safety-critical constraints are enforced by discrete-time” CBFs, and the future planning of MPC allows for better controls.

References

- [1] L. Wang, A. Ames, and M. Egerstedt, “Safety barrier certificates for heterogeneous multi-robot systems,” in *2016 American Control Conference (ACC)*, pp. 5213–5218, 2016.
- [2] J. Zeng, B. Zhang, and K. Sreenath, “Safety-critical model predictive control with discrete-time control barrier function,” 2021.